MECH 45X Task Planning

Airbus Robotics Flip Table

Group: Team 02

Date: March 12, 2026 | Week 10 Submission

1. Schedule Overview

This document covers the current state of the project as of Week 10. At this point, Dossiers 1 through 8 have been completed. Dossier 9 (Design Review) is nearly done, with only the updated budget still being finalized. The bulk of the remaining work falls under Dossier 10, which covers the detailed prototype design and the physical build itself.

The working prototype is targeted for April 7, 2026. Verification and validation (Dossier 11) runs from late March through early April, and project wrap-up (Dossier 12) follows immediately after.

The table below gives a high-level view of where things stand.

Phase	Dates	Status	Key Owner(s)
Dossier 9: Design Review	Jan 8 - Jan 23	Nearly Complete	Jordan, Kenneth, Oliver
Dossier 10: Detailed Prototype	Jan 8 - Apr 7	In Progress	All
Dossier 11: Verification	Mar 23 - Apr 5	Prep Started	All
Dossier 12: Project Wrap-Up	Apr 4 - Apr 13	Not Started	All

It is worth noting that Dossier 11 preparation has already started even though the prototype is not yet complete. This was a deliberate decision to front-load the documentation and scripting work so that only the actual test execution depends on having the hardware ready.

2. Current Progress

2.1 Dossier 9: Design Review

Dossier 9 covers the risk assessment, safety planning, and updated budget. Two of the three sub-tasks are complete:

- Risk Assessment (9A): Complete. Jordan led this from January 8 to 14. The assessment identified the key risks around supplier lead times and build window compression, which informed the contingency planning described later in this document.
- Safety Planning (9B): Complete. Kenneth handled this from January 16 to 21. The safety plan covers both the prototype itself and the testing procedures we will use during verification.
- Updated Budget (9C): 80% complete. Oliver has been working on this from January 8 to 23. The main thing holding this up is that we are still waiting on the final invoice for the remaining motor and gearbox delivery, so the procurement costs are not fully reconciled yet. Once that shipment arrives, the budget can be closed out.

2.2 Dossier 10: Detailed Prototype Design

This is the biggest and most active phase of the project right now. It covers everything from the design log and calculations through to the actual physical build. The table below shows the status of each sub-task.

Task	Owner	Progress	Dates	Notes
10A - Prototype Design Log	All	80%	Jan 23 - Mar 13	Updated weekly
Bill of Materials	All	80%	Jan 8 - Jan 30	
Procurement & Lead-Time Tracking	Oliver	90%	Jan 10 - Feb 10	Tracking ongoing
Updated Req. & Testing Matrix	Kenneth	80%	Jan 8 - Jan 16	
10B - Detailed Calculations	All	100%	Jan 12 - Jan 23	Complete
10C - CAD File of Final Prototype	Jordan, Mfon	90%	Jan 17 - Jan 30	Minor revisions left
10D - Controls Implementation	Gyan, Ryan	70%	Feb 10 - Mar 25	Parallel track
10E - Prototype Building	All	100%	Feb 10 - Mar 11	See breakdown

A few things worth highlighting:

- Detailed calculations (10B) are fully done. This includes all the structural analysis and motor sizing work that feeds into the CAD and the build.
- The CAD model (10C) is at 90%. Jordan and Mfon have been leading this. There are a few minor revisions left, mostly around mounting bracket geometry, but nothing that blocks the build.
- Controls implementation (10D) is at 70%. Gyan and Ryan have been developing the control logic in simulation. The software side is in good shape, but hardware integration cannot happen until the mechanical assembly is further along. This is expected to wrap up by March 25.
- The prototype build (10E) is the critical path item and is broken down in more detail in Section 4 below.

3. Procurement Status

Kenneth placed the parts order on January 22, immediately after the design was finalized and presented to Airbus. Oliver has been tracking lead times and delivery status since then. Here is where things stand:

- Guide rails, screws, and fasteners: These had a shorter lead time and have been fully received. Delivery window was February 5 through March 21, and everything came in within that range.

- Motors and gearboxes: This is the more challenging procurement item. The delivery window runs from February 19 through April 2. As of this submission, we have received about 50% of the order. The remaining units are expected by April 2.

The partial motor delivery is the single biggest schedule risk right now. We have structured the build plan (see next section) so that frame assembly and wiring can proceed in parallel without waiting for the full motor shipment. This means we are not losing time while we wait, but the mechanical integration phase (10E.2) cannot start until those parts arrive.

We have also contacted an alternative supplier as a backup in case the remaining delivery is delayed further. So far the original supplier has confirmed the April 2 date, but having a Plan B gives us some insurance.

4. Build Plan: Phased Breakdown

In the original Gantt chart, the prototype build (10E) was a single line item spanning several weeks. Based on feedback and to give better visibility into what is actually happening during the build, we have broken it down into four phases plus procurement milestones.

Build Sub-Phase	Owner	Progress	Dates	Dependencies / Notes
P1 - Order parts	Kenneth	100%	Jan 22	Complete
P2 - Delivery: rails, fasteners		100%	Feb 5 - Mar 21	All received
P3 - Delivery: motors, gearboxes		50%	Feb 19 - Apr 2	Partial; remainder due Apr 2
10E.1 - Frame & structure assembly	Oliver, Mfon	75%	Mar 15 - Mar 26	In progress
10E.2 - Mechanical integration	Gyan, Ryan	0%	Apr 2 - Apr 5	Blocked until motors arrive
10E.3 - Wiring & electrical	Ryan	30%	Mar 20 - Apr 6	Started early, in parallel
10E.4 - Testing + buffer	All	0%	Apr 1 - Apr 9	

Here is how the phases relate to each other:

- Phase 1 (Frame and structure assembly) started on March 15 and is currently 75% done. Oliver and Mfon are leading this. It uses the guide rails and fasteners that have already been received, so it was not blocked by the motor delivery situation.
- Phase 2 (Mechanical integration) is the phase that depends on the remaining motors and gearboxes. It is currently at 0% because we cannot start until those parts arrive around April 2. We have allocated 3 days for this (April 2 to 5), which is tight but achievable because Gyan and Ryan will be dedicated to it full-time, and all the mounting hardware and brackets are being prepared in advance.
- Phase 3 (Wiring and electrical) actually started early on March 20 at Ryan's initiative. Rather than waiting for the mechanical assembly to be fully done, Ryan began running cable harnesses and pre-wiring components that are already in place. This is at 30% and will continue through April 6.
- Phase 4 (Testing and buffer) runs from April 1 to 9. This overlaps slightly with the other phases because some functional tests can begin on sub-assemblies before the full prototype is integrated.

The controls work (10D) is a parallel track. Gyan and Ryan have been developing the control logic in simulation since February 10, and it is currently at 70%. The plan is to do the final hardware integration after the mechanical assembly (Phase 2) is complete. Because the software has been validated in simulation, we expect the hardware integration to go relatively smoothly.

The working prototype, prototype design evaluation, and prototype assessment memo are all targeted for April 7.

5. Verification and Validation Plan (Dossier 11)

One of the lessons from the feedback on the Week 2 submission was that the schedule left very little margin between the build and verification phases. To address this, we decided to start V&V preparation work on March 23, well before the prototype is expected to be complete. The idea is that most of the documentation, test procedures, and analysis scripts can be written without the hardware. Only the actual test execution requires a working prototype.

Task	Start	End	Notes
Iteration	Mar 23	Apr 5	Overall V&V window
Draft Verification Test Procedures	Mar 23	Mar 28	Before prototype is done
Dry-Run Verification Tests & Refinement	Mar 28	Apr 2	Using partial prototype
Data Processing & Analysis Scripts	Mar 23	Mar 28	Can be done without hardware
User Manual / Quick-Start & Maintenance	Mar 23	Mar 28	Based on CAD and design docs
11A - Verification (Eval of Specifications)	Apr 2	Apr 5	Requires working prototype
11B - Validation (User Evaluation)	Apr 5	Apr 5	Client-facing evaluation
Safety Validation & Verification Document	Apr 3	Apr 3	
Final Prototype Design Evaluations	Apr 4	Apr 4	

The iteration window (March 23 to April 5) is structured in two phases:

- Preparation (March 23 to 28): During this time, we will draft the verification test procedures, write the data processing and analysis scripts, and put together the user manual and quick-start guide. None of this requires the prototype to be finished, so it can happen in parallel with the build.
- Execution (March 28 to April 5): Dry-run tests start March 28 using whatever portion of the prototype is available. The formal verification (11A) runs April 2 to 5 once the full prototype is integrated. User evaluation (11B) is scheduled for April 5.

The safety validation and verification document is due April 3, and final prototype design evaluations happen April 4. These are relatively quick deliverables that build on the test results from 11A.

6. Project Wrap-Up (Dossier 12)

Dossier 12 covers all the final deliverables, presentations, and administrative closeout items. Most of the work here begins on April 4 and runs through April 13. The table below shows the full breakdown.

Task	Start	End	Notes
12A - Design and Innovation Day	Apr 9	Apr 9	Presentation day
12B - Final Report	Apr 4	Apr 12	
12C - Public Web Abstract	Apr 4	Apr 12	
12D - Electronic Archive	Apr 4	Apr 12	
12E - Logbook Submission	Apr 4	Apr 12	
12F - Personal Reflection	Apr 4	Apr 12	
12G - Client Review + Assessment Form	Apr 13	Apr 13	
12H - Client Check-Off Form	Apr 13	Apr 13	
12I - Financial Statement	Apr 7	Apr 13	
12J - Locker/Workspace Cleanout Form	Apr 10	Apr 13	
Final Documentation Completion	Apr 12	Apr 12	
Project Completion	Apr 12	Apr 12	

Design and Innovation Day is on April 9, which is the public-facing presentation of the project. The final report, web abstract, electronic archive, logbook, and personal reflections all have an April 4 to 12 window, giving us about a week to finalize everything after the prototype is demonstrated.

The client-facing items (12G and 12H) are scheduled for April 13, which is the last day we need client sign-off. The financial statement (12I) starts a bit earlier on April 7 since it depends on finalizing all procurement costs. The locker and workspace cleanout (12J) is a straightforward administrative task on the final days.

7. Risk and Contingency Planning

The table below summarizes the key risks we have identified and the mitigation strategies for each. The most critical risk is the remaining motor and gearbox delivery. Everything downstream of that (mechanical integration, testing, and to some extent V&V) is affected if those parts do not arrive on time.

Risk	Likelihood	Impact	Mitigation
Remaining motors/gearboxes delayed past Apr 2	High	High	50% already received. Alt supplier contacted as backup. Frame and wiring continue independently so we are not sitting idle.
Mechanical integration window too tight (3 days)	High	High	Pre-plan all integration steps and prepare mounting hardware before motors arrive. Assign dedicated team (Gyan, Ryan) for those 3 days.

Controls integration issues during final assembly	Medium	High	Controls at 70% in simulation. Most logic is validated before touching hardware. Ryan doing wiring in parallel to reduce integration unknowns.
Testing window too short to cover all specs	Medium	High	Prioritize critical function tests first. Test procedures and data scripts written in advance (Mar 23-28) so execution is efficient.
Prototype not ready for Apr 7 milestone	Medium	High	Parallel workstreams reduce serial dependency. Worst case: demonstrate with partial integration and document remaining items.
V&V compressed by build delays	Medium	Medium	V&V prep (procedures, scripts, user manual) starts Mar 23 before prototype is done. Only the actual test execution needs the hardware.
Team availability conflicts (coursework)	Low	Medium	Midterm and assignment dates mapped against build schedule. Tasks distributed so no single person is a bottleneck.

The overall approach to risk mitigation is built around parallel workstreams. Rather than having a purely sequential build process where each phase waits for the previous one, we have structured things so that frame assembly, wiring, controls development, and V&V preparation all proceed at the same time. This means that a delay in one area (like motor delivery) does not cause everything else to stop.

The other key mitigation is advance preparation. For the mechanical integration phase, which is the most time-constrained (3 days), Gyan and Ryan are preparing all the mounting hardware, brackets, and tooling ahead of time so that when the motors arrive, they can go straight into installation without any setup delays.

8. Accelerated Schedule Justification

The project timeline is tighter than a typical capstone project, and this is a direct result of two factors:

- Client requirements: Airbus asked for a detailed CAD and design solution to be presented by Week 3 (January 21). This meant the design phase was compressed, but it also meant we could start procurement earlier than most teams. Kenneth placed the parts order on January 22, the day after the client presentation.
- Supplier lead times: The best-case lead time for our motors and gearboxes is 4 weeks, and worst case is 8+ weeks. For a product at the scale of the flip table, these are standard industrial lead times, but they eat into the build window significantly. The decision to order early and structure the build around parallel workstreams was specifically to absorb this variability.

Without the parallel workstream approach, the project would need an additional 3 to 4 weeks of serial execution time. By having the frame, wiring, controls, and V&V prep all running simultaneously, we are able to fit everything within the available window while still maintaining a reasonable level of schedule resilience.

9. Next Steps

Looking at the next two to three weeks, the key priorities are:

- Finish frame assembly by March 26 (Oliver, Mfon).
- Continue wiring work in parallel, targeting completion by April 6 (Ryan).
- Complete V&V preparation documents by March 28 (test procedures, data scripts, user manual).
- Receive remaining motors and gearboxes and begin mechanical integration around April 2 (Gyan, Ryan).
- Finalize controls hardware integration after mechanical assembly is done.
- Hit the working prototype milestone on April 7.
- Start Dossier 12 wrap-up tasks on April 4.
- Final documentation and project completion by April 12 to 13.

The Gantt chart will continue to be updated weekly with actual completion rates to track progress against this plan.